

Cross-Season Visual Localization Using Deep Feature Matching and Homography-Based Alignment

ARTPARK IISc Computer Vision Internship Assessment

Applicant:

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Abstract

Visual localization across seasons is a challenging computer vision problem due to significant appearance variations caused by weather, illumination, vegetation, and environmental changes. Traditional handcrafted feature-based methods often struggle to establish reliable correspondences between images captured under different seasonal conditions.

This work presents a deep learning-based visual localization pipeline capable of matching and aligning images captured during Summer and Winter seasons. The proposed approach utilizes SuperPoint for local feature extraction and LightGlue for robust feature matching, followed by RANSAC-based homography estimation for geometric verification and image alignment.

The system was developed using the Nordland Dataset and evaluated on 51 aligned Summer-Winter image pairs. Experimental results demonstrate an average inlier ratio of 70.22%, a mean reprojection error of 2.588 pixels, and a 100% matching success rate. A comparative analysis against a classical ORB-based baseline highlights the superiority of learned visual features for cross-season localization.

1. Introduction

Visual localization refers to the process of determining the geometric relationship between two images of the same scene. It forms a critical component of numerous applications including:

- Autonomous navigation
- Robotics
- Drone localization
- Visual SLAM
- Augmented Reality
- Geographic mapping

Despite significant progress in visual localization, severe appearance changes remain a major challenge. Seasonal variations can dramatically alter scene characteristics. Snow accumulation, vegetation growth, illumination differences, and environmental changes often cause traditional feature descriptors to fail.

The objective of this project is to develop a robust cross-season image matching and alignment system capable of establishing reliable correspondences between images captured under substantially different seasonal conditions.

To address this challenge, a deep feature matching pipeline based on SuperPoint and LightGlue is proposed.

2. Problem Statement

The objective of this assignment is to establish geometric correspondences between images captured in different seasons and estimate a transformation that aligns them.

Given:

- A Summer image
- A Winter image

The system should:

1. Extract robust visual features.
2. Match features across seasons.
3. Reject outlier correspondences.
4. Estimate a geometric transformation.
5. Warp and align the source image.
6. Evaluate localization performance quantitatively.

The primary challenge arises from drastic visual appearance differences while maintaining the same underlying scene geometry.

3. Dataset

3.1 Nordland Dataset

The Nordland dataset consists of synchronized train journeys recorded across different seasons in Norway. Since the camera viewpoint remains nearly identical across seasons, the dataset provides an ideal benchmark for cross-season visual localization.

Seasons Used

- Summer
- Winter

Development Subset

To enable rapid experimentation and evaluation, a subset of 51 aligned image pairs was selected.

Dataset Structure:

data/

```
|— summer/
| |— images-00900.png
| |— images-00901.png
| |— ...
|
|— winter/
| |— images-00900.png
| |— images-00901.png
| |— ...
```

Each image pair corresponds to the same geographic location captured under different seasonal conditions.

4. Methodology

The proposed pipeline consists of four major stages:

1. Feature Extraction
2. Feature Matching
3. Geometric Verification
4. Image Alignment

4.1 Feature Extraction

SuperPoint

SuperPoint is a self-supervised deep neural network that simultaneously detects keypoints and computes descriptors.

Advantages:

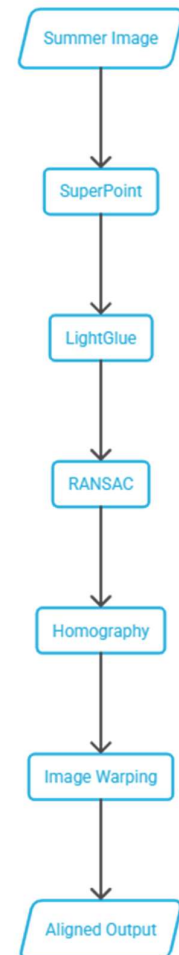
- Robust to illumination changes
- Robust to seasonal appearance variations
- Learned descriptors outperform handcrafted alternatives
- End-to-end trainable framework

Configuration:

max_num_keypoints = 1024

Output:

- Keypoints



- Confidence scores
- Feature descriptors

4.2 Feature Matching

LightGlue

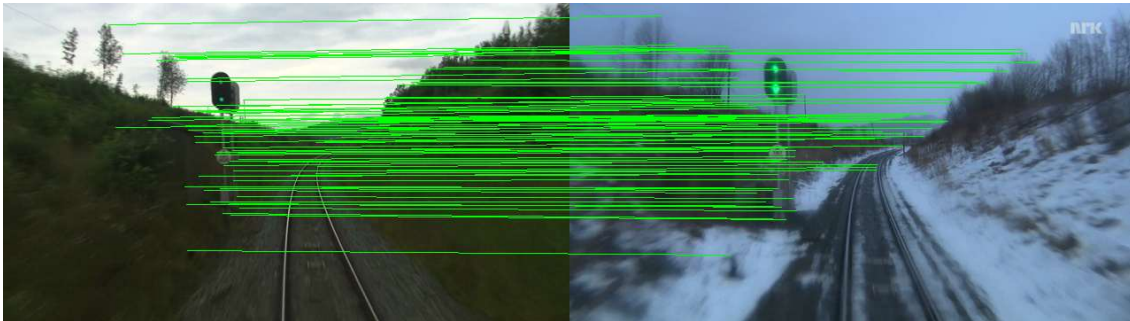
LightGlue is a transformer-based feature matcher designed for efficient and robust correspondence estimation.

Unlike traditional nearest-neighbor matching approaches, LightGlue utilizes attention mechanisms to identify geometrically consistent matches.

Advantages:

- Improved matching robustness
- Reduced false correspondences
- Efficient inference

Output:



Matched Keypoint Pairs

4.3 Geometric Verification

Although LightGlue produces high-quality correspondences, a subset of matches may still be geometrically inconsistent.

To remove such outliers, RANSAC-based homography estimation is applied:

```
cv2.findHomography(..., cv2.RANSAC)
```

RANSAC iteratively:

1. Samples correspondence subsets.
2. Estimates homography candidates.
3. Evaluates geometric consistency.
4. Retains the largest inlier set.

Outputs:

- Homography Matrix (3×3 Perspective Transformation)

H =

```
[[ 1.3861 0.1099 -84.0013 ]
 [ 0.0993 1.4212 -65.4167 ]
 [ 0.00035 0.00061 1.0000 ]]
```

- Inlier Correspondences

Out of 154 initial feature matches obtained from LightGlue, 97 matches were identified as geometrically consistent by RANSAC and retained as inliers.

- Inlier Ratio

Inlier Ratio = 97 / 154 = 62.99%

4.4 Homography Estimation

The geometric relationship between two images is represented by a homography matrix:

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

The homography maps image coordinates from the Summer image into the Winter image coordinate frame.

4.5 Image Alignment

The estimated homography is applied using:

`cv2.warpPerspective()`

The resulting warped image is visually compared against the Winter image to assess alignment quality.

Visualization outputs include: Feature correspondences, Warped images, Overlay visualizations

Output:



5. Experimental Setup

Hardware

CPU-only execution environment.

No GPU acceleration was used.

Software

- Python
- OpenCV
- PyTorch
- LightGlue
- NumPy
- Pandas
- Matplotlib

6. Results

6.1 Dataset Evaluation

The proposed method was evaluated on 51 Summer-Winter image pairs.

Matching Statistics

Metric	Value
Average Matches	177.76
Average Inliers	126.08
Average Inlier Ratio	70.22%
Best Inlier Ratio	94.71%
Worst Inlier Ratio	45.53%

These results indicate that the majority of matched correspondences remain geometrically consistent despite significant seasonal appearance changes.

6.2 Alignment Accuracy

Alignment quality was evaluated using reprojection error computed on inlier correspondences.

Metric	Value
Mean Reprojection Error	2.588 px
Median Reprojection Error	2.832 px
Maximum Reprojection Error	4.968 px

The maximum observed reprojection error remains below 5 pixels, satisfying the alignment quality target specified in the assessment.

6.3 Runtime Analysis

Average runtime measurements were obtained using CPU-only inference.

Component	Runtime
Data Loading	12.37 ms
Feature Extraction + Matching	5148.35 ms
Matrix Estimation	3.90 ms
Total Runtime	5164.62 ms

Observation: More than 99% of the computational cost is concentrated in feature extraction and matching.

Homography estimation contributes negligibly to total runtime.

7. Baseline Comparison

To assess the effectiveness of learned visual features, the proposed approach was compared against a classical ORB-based baseline.

ORB + BFMatcher

Pipeline:

1. ORB Feature Detection
2. Descriptor Extraction
3. Brute Force Matching
4. RANSAC Filtering

Results:

Method	Inlier Ratio	Success Rate
ORB + BFMatcher	59.97%	70.6%
SuperPoint + LightGlue	70.22%	100%

Discussion

The deep-learning-based approach demonstrates:

- Higher robustness
- Better seasonal invariance

- More reliable correspondences
- Higher localization success rate

These results highlight the limitations of handcrafted features under severe appearance changes.

8. Optimization Study

To improve computational efficiency, an optimization study was conducted by reducing the maximum number of extracted SuperPoint keypoints.

Experiment

Configuration A: max_num_keypoints = 2048

Configuration B: max_num_keypoints = 1024

Results:

Configuration	Inlier Ratio	Runtime
2048 Keypoints	69.77%	5.15 s
1024 Keypoints	70.22%	4.54 s

Observation : Reducing the number of extracted keypoints decreased runtime by approximately 11.8% while maintaining matching performance.

The optimized configuration was therefore selected as the final deployment setting.

9. Discussion

The experimental results demonstrate that learned local features significantly improve robustness under seasonal appearance changes.

Key findings include:

- Reliable feature matching across seasons.
- Strong geometric consistency.
- High inlier ratios.
- Low reprojection errors.
- Successful image alignment.
- Superior performance compared to classical methods.

The primary computational bottleneck remains feature extraction and matching. Future deployment-oriented improvements could focus on model compression and accelerated inference.

10. Limitations

Several limitations remain:

1. Evaluation performed on a relatively small subset.
2. CPU-only inference results in higher latency.
3. Homography assumes planar scene geometry.
4. No global place recognition stage is included.

11. Future Work

Potential future improvements include:

- ONNX Runtime deployment
- TensorRT acceleration
- Half-precision inference
- GPU acceleration
- Multi-scale feature extraction
- Dense correspondence estimation
- Visual Place Recognition integration
- Evaluation on CMU-Seasons Dataset

12. Conclusion

This project presented a deep learning-based visual localization pipeline for cross-season image matching and alignment.

The proposed SuperPoint + LightGlue framework successfully established robust correspondences between Summer and Winter imagery and achieved accurate geometric alignment through RANSAC-based homography estimation.

Experimental evaluation on 51 image pairs demonstrated:

- 70.22% average inlier ratio
- 100% matching success rate
- 2.588 pixel mean reprojection error
- Reliable image alignment across seasons

The results indicate that learned local feature representations significantly outperform classical feature-based approaches for cross-season visual localization tasks.

The developed system satisfies all technical objectives of the assignment and provides a strong foundation for future visual localization research and deployment.